

TSEGAY TESFAY MEZGEBE, PhD

Industrial & Systems Engineering | Manufacturing Automation | Lean & Optimization | R&D Leadership

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PROFESSIONAL SUMMARY

PhD-qualified industrial and systems engineer with 17+ years across academia and industry, specializing in manufacturing automation, intelligent/Industry 4.0 systems, lean manufacturing, and optimization. Proven record leading multi-million-euro international projects, delivering high-value industrial feasibility studies and consultancy across manufacturing, metals, pharmaceuticals, and infrastructure, and driving R&D and technology transfer. Skilled in multi-agent control, process modeling, and data-driven decision-making, with strong leadership experience directing engineering schools, R&D units, and international partnerships.

CORE COMPETENCIES

Manufacturing Automation & Control • Lean Manufacturing & Six Sigma • Industry 4.0 / Intelligent Manufacturing • Optimization & Meta-Heuristics • Feasibility Studies & Investment Analysis • Quality Management (ISO 9001, ISO 19011) • Multi-Agent Systems • Process & Systems Modeling • Project & R&D Leadership • Data Analysis (Minitab, SPSS, MATLAB)

PROFESSIONAL EXPERIENCE

Director, International Affairs Directorate — Mekelle University, Ethiopia *Jun 2026 – Present*

Lead international partnerships, mobility programs, and institutional collaboration strategy.

Director, R&D — Beles Tech Research and Development, Addis Ababa, Ethiopia *Nov 2024 – Jun 2026*

Directed research and development operations for prickly pear cactus value-chain innovation.

Dean, School of Mechanical & Industrial Engineering — EiT-M, Mekelle University *Jun 2021 – Oct 2025*

Led academic strategy, staffing, and operations for a major engineering school; drove curriculum modernization and industry linkage.

Associate Professor of Industrial & Systems Engineering — EiT-M, Mekelle University *Jan 2025 – Present*

Teach and supervise graduate research in automation, manufacturing control, and optimization; lead funded projects and industrial consultancy.

Assistant Professor & Head of Postgraduate Studies — EiT-M, Mekelle University *2020 – 2025*

Directed postgraduate programs and quality assurance; supervised doctoral and masters research.

Postdoctoral Researcher — CRAN, University of Lorraine, France *Dec 2019 – Mar 2020*

Researched distributed manufacturing control using multi-agent systems in collaboration with international partners.

Acting Head, Merchandizing Division / Merchandizer — MAA Garment & Textile Factory, Ethiopia *2008*

Managed merchandizing operations and monitored garment production activities on the plant floor.

CONSULTANCY & INDUSTRIAL PROJECTS (SELECTED)

- Led HR job-grading and salary-scale restructuring for Mesfin Metals & Engineering PLC (ETB 450,000).
- Directed maintenance inspection and diagnostics of heavy-duty machinery for the Tigray Roads Authority (ETB 1.9 million).
- Delivered feasibility studies for a galvanized steel factory and a metal-casting technology plant (Mesfin Industrial Engineering).
- Led feasibility study for an Olive Pharmaceutical manufacturing plant, hospital, and cardiac center (ETB 972,900).
- Local Coordinator / Project Manager for the €1.39M EU-funded SPREE engineering-education mobility program across Africa.

- Environmental Management Plan study for Messebo Cement Factory (ETB 1.2 million).

EDUCATION

PhD in Automatic Control, CRAN, University of Lorraine, France (awarded January 15, 2020) 2017 – 2020
 MSc in Production and Industrial Systems Engineering, Ethiopian Institute of Technology-Mekelle, Mekelle University, Ethiopia (awarded October 27, 2011) 2009 – 2011
 BSc in Industrial Engineering, Faculty of Engineering, Bahir Dar University, Ethiopia (awarded June 06, 2007) 2003 – 2007

CERTIFICATIONS & TRAINING (SELECTED)

Seminar on “Poverty Reduction and Alleviation through Business and Trade for Developing Countries” – Ministry of Commerce of China & China Center for Urban Development (NDRC), Beijing, China May–Jun 2026
 Artificial Intelligence Fundamentals – Udacity, USA Oct–Nov 2024
 Data Analysis Fundamentals – Udacity, USA Sep 2024
 Meta-Heuristics in Optimization: Current Trends and Evolutions – Ecole MACS 2019, University of Bordeaux, France Jun 2019
 Quality Management System Development & Implementation (ISO 9001:2008) – Ethiopian Standards Agency Mar 2015
 Quality Management Systems Audit (ISO 19011:2011) – Ethiopian Standards Agency Apr 2015

SKILLS

Software & Engineering: SCADA, MATLAB, Python, Java, SolidWorks, TeX/LaTeX
Statistical & Analysis: Minitab, MATLAB, ClickCharts, SPSS
Languages: Tigrigna (native), English (fluent), Amharic (fluent), French (working proficiency)

SELECTED PUBLICATIONS & RECOGNITION

Author of 20+ peer-reviewed publications in journals including Journal of Intelligent Manufacturing, Scientific Reports, Computers in Industry, and Scientific African, plus a doctoral dissertation on human-inspired control algorithms for the Factory of the Future. Honorary Adjunct Professor, Chandigarh Group of Colleges, India. Full list available on Google Scholar and ORCID.

REFERENCES

- Dr. Mulualem Gebregiorgis Gebreslassie – Associate Professor of Energy, Sheffield Hallam University, UK. M.Gebreslassie@shu.ac.uk
- Prof. Hind Bril El-Haouzi – Full Professor of Industrial Engineering, CRAN, University of Lorraine, France. hind.el-haouzi@univ-lorraine.fr
- Dr. Alula Gebresas – Associate Professor of Materials Science and Technology, EiT-M, Mekelle University, Ethiopia; Postdoctoral Researcher, Technical University of Clausthal, Germany. alula.gebresas@tu-clausthal.de